

Thangal Kunju Musaliar College of Engineering

(Government Aided and Autonomous)

Kollam, Kerala



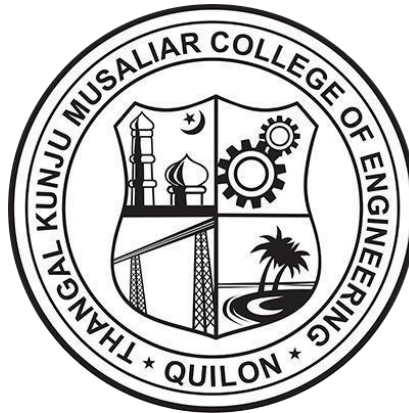
IDEA LAB PROJECT REPORT

Department of Electrical and Electronics Engineering

December 2025

Thangal Kunju Musaliar College of Engineering
(Government Aided and Autonomous)

Kollam, Kerala



IDEA LAB PROJECT REPORT

CERTIFICATE

*Certified that this is a bonafide record of the group project work done by **AVIGHNA KRISHNA R, AYISHA KOCHUMANGALAM, ASHTAMI S CHAND, JANAKI JAYIR, PRAYAG KRISHNAN U** of **FIRST** Semester, Class **E1.A** Roll Nos.75,24,21,82,52 in the **IDEA Lab** during the year **2025-26***

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Staff member in charge

VISION

To be a centre of excellence imparting quality education in Computer Science and Engineering and transforming students to critical thinkers and lifelong learners capable of developing environment friendly and economically feasible solutions to real world problems.

MISSION

- To provide strong foundation in Computer Science and Engineering, prepare students for professional career and higher education, and inculcate research interest.
- To be abreast of the technological advances in a rapidly changing world.
- To impart skills to come up with socially acceptable solutions to real world problems, upholding ethical values.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOS)

PEO 1: Excel in professional career by acquiring knowledge in mathematics, science, and engineering and applying the knowledge in the design of hardware and software Solutions for challenging problems of the society.

PEO 2: Pursue higher studies and research thereby engages in lifelong learning by adapting to the current trends in the area of Computer Science & Engineering.

PEO 3: Ability to provide socially acceptable and economically feasible computer oriented solutions to real world problems with teamwork, while maintaining environmental balance, quality and cognizance of the underlying principles of ethics.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Apply mathematical and algorithmic principles, data structure concepts, software, and hardware and techniques in designing and developing optimized and secure computer based solutions.

PSO2: Design and develop system software and provide exposure to various tools and programming languages to facilitate efficient computing environment which adds to the ease of human life.

PSO3: Use the knowledge of various data processing, communication and intelligent systems to provide solutions to new ideas and innovations.

PROGRAMME OUTCOMES

1. Engineering knowledge : Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. Problem analysis : Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first Principles of mathematics, natural sciences, and engineering sciences.
3. Design/development of solutions : Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. Conduct investigations of complex problems : Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. The engineer and society : Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. Environment and sustainability : Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. Ethics : Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. Individual and team work : Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10. Communication : Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. Project management and finance : Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. Life-long learning : Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

COURSE OUTCOMES (COs)

CO1	Develop project using appropriate Microcontroller Programming languages. [Apply level]
CO2	Develop product using PCB Design and Prototyping concepts. [Apply level]
CO3	Create 2D and 3D models using appropriate tools. [Apply level]
CO4	Create electronic documentation for the system/project using appropriate tools. [Apply level]
CO5	Build useful and standalone system/ project with enclosures. [Apply level]

BLUETOOTH CONTROLLED CAR USING ARDUINO UNO

ACKNOWLEDGEMENT

We would like to thank our Project Supervisor, Prof. Ruksana, for her invaluable guidance, constant support, and constructive feedback throughout the course of this project.

We also wish to extend sincere thanks to our Head of Department, Dr. Sheeba.R, and our Senior Advisor, Dr. Resmi.R , for their insights and for providing the necessary resources that made this project possible.

ABSTRACT

The project “Bluetooth Controlled Car Using Arduino” is designed to control a small car wirelessly using a smartphone. It works with an Arduino board, a Bluetooth module (HC- 05), and an L298N motor driver to control the direction and movement of the car. The user sends commands from a mobile app through Bluetooth, which are received by the Arduino. The Arduino then drives the motors using the motor driver to move the car forward, backward, left, right, or stop. The car is powered by a battery, making it portable and easy to use. This project helps to understand how Bluetooth communication and motor control work with Arduino. It is a simple and low-cost system that can be used for basic robotics learning and can later be upgraded with sensors for obstacle detection or automation. Overall, the project shows how wireless control can be applied in small robotic systems.

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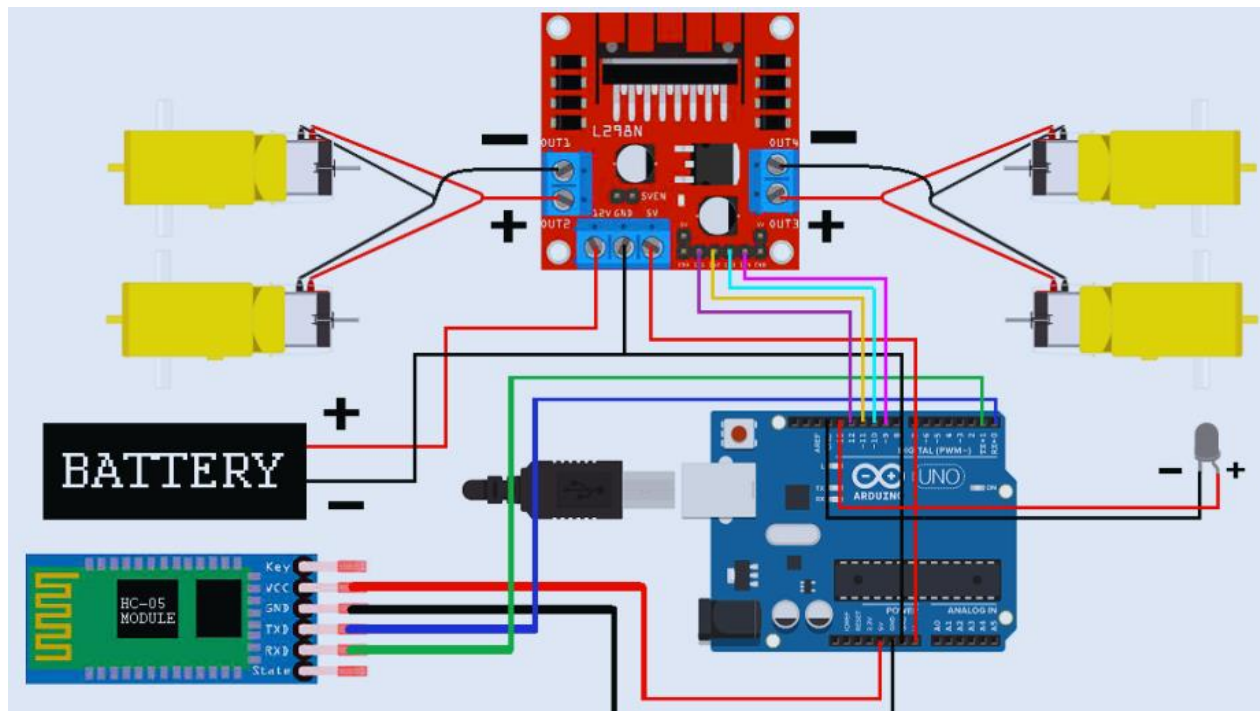
INTRODUCTION

The Bluetooth Controlled Arduino Car is a simple and efficient wireless robotic system designed to demonstrate the use of Bluetooth communication in embedded electronics. The project uses an Arduino UNO as the main controller, an HC-05 Bluetooth module for wireless connectivity, and an L298N motor driver to operate the DC motors. By sending commands from a smartphone application, the user can easily control the movement of the car in different directions such as forward, backward, left, right, and stop. This project introduces the basics of robotics, wireless control, and hardware–software interfacing. It also provides a hands-on understanding of how microcontrollers and sensors work together to create smart automated systems.

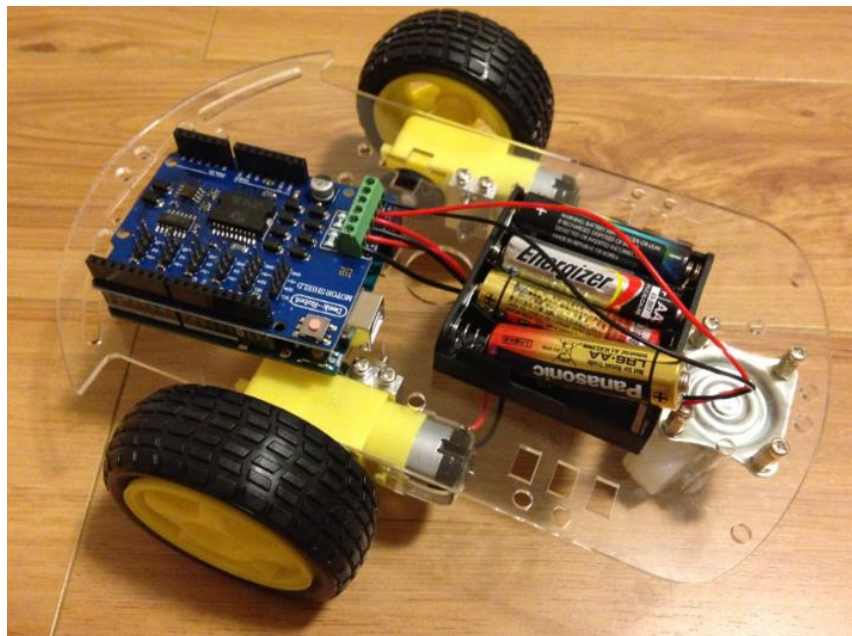
MOTIVATION

The primary motivation behind this project is to learn and explore the fundamentals of wireless technology and robotics in a practical way. As Bluetooth communication is widely used in modern electronic devices, building a Bluetooth-controlled car helps students understand how real-world wireless systems function. The project also aims to encourage creativity and problem-solving by giving learners the opportunity to design, assemble, and program their own robotic vehicle. Additionally, this model can later be expanded with sensors and IoT technologies, inspiring students to develop advanced and innovative automation projects in the future.

DESIGN



TINKER CAD CIRCUIT



BLUETOOTH ARDUINO CAR

IMPLEMENTATION

```

#include <SoftwareSerial.h>

// Bluetooth module pins
const int BT_RX = 2; // Arduino RX (connect to HC-05 TX)
const int BT_TX = 3; // Arduino TX (connect to HC-05 RX via voltage divider)

// Initialize SoftwareSerial
SoftwareSerial Bluetooth(BT_RX, BT_TX);

// Motor Pins
const int LEFT_IN1 = 9;
const int LEFT_IN2 = 10;
const int LEFT_EN = 5;

const int RIGHT_IN1 = 11;
const int RIGHT_IN2 = 12;
const int RIGHT_EN = 6;

int speed = 200;
char t;

void setup() {
  // Motor pins
  pinMode(LEFT_IN1, OUTPUT);
  pinMode(LEFT_IN2, OUTPUT);
  pinMode(LEFT_EN, OUTPUT);

  pinMode(RIGHT_IN1, OUTPUT);
  pinMode(RIGHT_IN2, OUTPUT);
  pinMode(RIGHT_EN, OUTPUT);

  // Start Bluetooth serial
  Bluetooth.begin(9600);
}

void loop() {
  if (Bluetooth.available() > 0) {
    t = Bluetooth.read();

    switch(t) {
      case 'F': forward(); break;
      case 'B': backward(); break;
      case 'L': leftward(); break;
      case 'R': rightward(); break;
    }
  }
}

```

```

    case 'S': stop_the(); break;
  }
}
}

// Motor functions
void forward() {
  analogWrite(LEFT_EN, speed);
  analogWrite(RIGHT_EN, speed);

  digitalWrite(LEFT_IN1, HIGH);
  digitalWrite(LEFT_IN2, LOW);
  digitalWrite(RIGHT_IN1, HIGH);
  digitalWrite(RIGHT_IN2, LOW);
}

void backward() {
  analogWrite(LEFT_EN, speed);
  analogWrite(RIGHT_EN, speed);

  digitalWrite(LEFT_IN1, LOW);
  digitalWrite(LEFT_IN2, HIGH);
  digitalWrite(RIGHT_IN1, LOW);
  digitalWrite(RIGHT_IN2, HIGH);
}

void stop_the() {
  analogWrite(LEFT_EN, 0);
  analogWrite(RIGHT_EN, 0);

  digitalWrite(LEFT_IN1, LOW);
  digitalWrite(LEFT_IN2, LOW);
  digitalWrite(RIGHT_IN1, LOW);
  digitalWrite(RIGHT_IN2, LOW);
}

void leftward() {
  analogWrite(LEFT_EN, speed / 2);
  analogWrite(RIGHT_EN, speed);

  digitalWrite(LEFT_IN1, HIGH);
  digitalWrite(LEFT_IN2, LOW);
  digitalWrite(RIGHT_IN1, HIGH);
  digitalWrite(RIGHT_IN2, LOW);
}

void rightward() {
  analogWrite(LEFT_EN, speed);

```

```
analogWrite(RIGHT_EN, speed / 2);  
  
digitalWrite(LEFT_IN1, HIGH);  
digitalWrite(LEFT_IN2, LOW);  
digitalWrite(RIGHT_IN1, HIGH);  
digitalWrite(RIGHT_IN2, LOW);  
}
```


bluetoothcar.c File Reference

```
#include <SoftwareSerial.h>
```

Functions

```
SoftwareSerial Bluetooth (BT_RX, BT_TX)
```

SoftwareSerial object for Bluetooth communication.

```
void setup ()
```

Arduino initialization.

```
void loop ()
```

Main execution loop.

```
void forward ()
```

Drive both motors forward.

```
void backward ()
```

Drive both motors backward.

```
void stop_the ()
```

Stop both motors completely.

```
void leftward ()
```

Turn left by slowing the left motor and running the right motor at full speed.

```
void rightward ()
```

Turn right by slowing the right motor and running the left motor at full speed.

Variables

```
const int BT_RX = 2
```

Bluetooth module RX pin (Arduino reads from this pin). Connect HC-05 → TX.

```
const int BT_TX = 3
```

Bluetooth module TX pin (Arduino writes to this pin). Connect HC-05 → RX (with voltage divider).

```
const int LEFT_IN1 = 9
```

Left motor input pin 1.

```
const int LEFT_IN2 = 10
```

Left motor input pin 2.

```
const int LEFT_EN = 5
```

Left motor enable pin (PWM)

```
const int RIGHT_IN1 = 11
```

Right motor input pin 1.

```
const int RIGHT_IN2 = 12
```

Right motor input pin 2.

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bluetooth car: bluetoothcar.c File Reference

<code>const int RIGHT_EN = 6</code>
Right motor enable pin (PWM)
<code>int speed = 200</code>
Default motor speed (0–255).
<code>char t</code>
Stores incoming Bluetooth command character.

Function Documentation

◆ backward()

```
void backward ( )
```

Drive both motors backward.

◆ Bluetooth()

```
SoftwareSerial Bluetooth ( BT_RX ,  
                          BT_TX )
```

SoftwareSerial object for Bluetooth communication.

◆ forward()

```
void forward ( )
```

Drive both motors forward.

◆ leftward()

```
void leftward ( )
```

Turn left by slowing the left motor and running the right motor at full speed.

◆ loop()

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bluetooth car: bluetoothcar.c File Reference

```
void loop ( )
```

Main execution loop.

Reads characters from Bluetooth and calls the corresponding motor control function.

◆ rightward()

```
void rightward ( )
```

Turn right by slowing the right motor and running the left motor at full speed.

◆ setup()

```
void setup ( )
```

Arduino initialization.

Sets pin modes and starts Bluetooth serial communication.

◆ stop_the()

```
void stop_the ( )
```

Stop both motors completely.

Variable Documentation

◆ BT_RX

```
const int BT_RX = 2
```

Bluetooth module RX pin (Arduino reads from this pin). Connect HC-05 → TX.

◆ BT_TX

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bluetooth car: bluetoothcar.c File Reference

```
const int BT_TX = 3
```

Bluetooth module TX pin (Arduino writes to this pin). Connect HC-05 → RX (with voltage divider).

◆ LEFT_EN

```
const int LEFT_EN = 5
```

Left motor enable pin (PWM)

◆ LEFT_IN1

```
const int LEFT_IN1 = 9
```

Left motor input pin 1.

◆ LEFT_IN2

```
const int LEFT_IN2 = 10
```

Left motor input pin 2.

◆ RIGHT_EN

```
const int RIGHT_EN = 6
```

Right motor enable pin (PWM)

◆ RIGHT_IN1

```
const int RIGHT_IN1 = 11
```

Right motor input pin 1.

◆ RIGHT_IN2

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bluetooth car: bluetoothcar.c File Reference

```
const int RIGHT_IN2 = 12
```

Right motor input pin 2.

◆ speed

```
int speed = 200
```

Default motor speed (0–255).

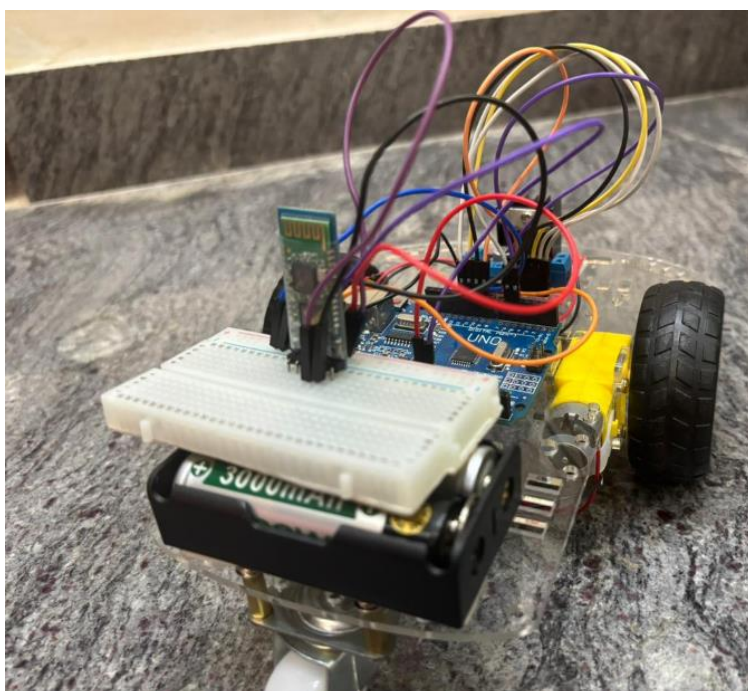
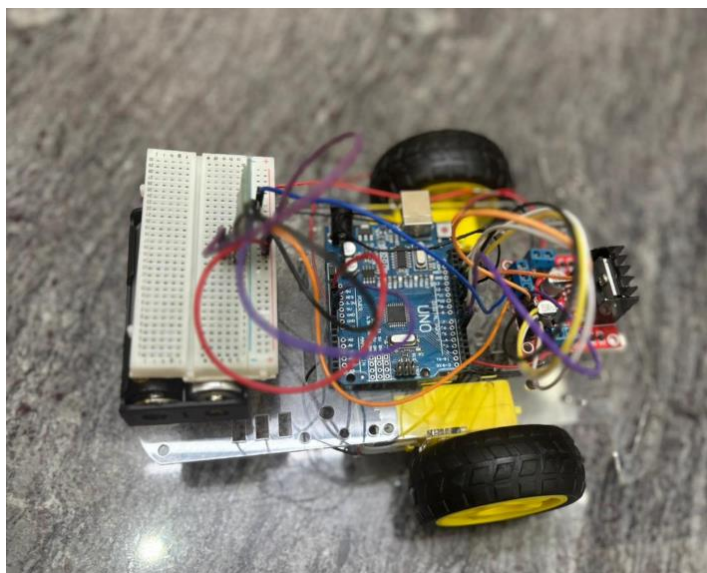
◆ t

```
char t
```

Stores incoming Bluetooth command character.

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RESULT



CONCLUSION

- Demonstrates wireless control using Bluetooth communication.
- Integrates Arduino, HC-05 Bluetooth module, and L298N motor driver effectively.
- Shows how Bluetooth enables short-range wireless operation.
- Provides practical understanding of hardware interfacing, motor control, and serial communication.
- Enhances skills in electronics, embedded programming, and robotics.
- Acts as a foundation for advanced systems like:
 - IoT-based vehicles
 - Obstacle-avoiding robots
 - Smart mobility and automation solutions

FUTURE SCOPES

1. Obstacle Detection and Avoidance

Adding ultrasonic or infrared sensors can help the car detect obstacles and automatically avoid collisions, making it more autonomous.

2. IoT-Based Control

By replacing Bluetooth with Wi-Fi or IoT modules (ESP8266/ESP32), the car can be controlled from anywhere in the world using an internet-based app.

3. Line-Following and Path Planning

Sensors such as IR line sensors can be added to make the car follow predefined paths or routes, useful for automation and navigation studies.

4. Voice Control or Gesture Control

Integrating voice recognition (using apps or modules) or hand gesture sensors can enable more advanced and user-friendly control methods.

5. Camera Integration (FPV Control)

A small wireless camera can be mounted on the car to enable live video streaming, allowing the user to control the car like a remote-controlled surveillance robot.

6. Solar or Rechargeable Power System

Replacing normal batteries with rechargeable or solar-powered systems can improve efficiency and reduce running costs.

7. Enhanced Speed and Power

Using stronger motors and a better chassis can make the car suitable for rough terrain or higher-speed applications.

8. Autonomous Navigation Using AI

With additional sensors and programming, the car can be upgraded to navigate without human control, similar to self-driving technology.

REFERENCES

- GitHub <https://github.com>
- Arduino.cc <https://www.arduino.cc>

APPENDIX

Component specifications:-

- Arduino UNO
- HC-05 Bluetooth Module
- L298N Motor Driver DC
- DC Motors (2 or 4)
- Chassis & Wheels
- Battery
- Jumper Wires
- Smartphone App (e.g., Bluetooth RC Controller)

